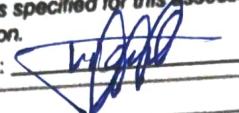


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2- Concept Mapping

Course Code:	CSA0311	Course Title:	Data Structures
CO Assessed:	CO1 - Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 - Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	Vignesh P.S	Reg. No.:	192511177
Section / Batch:		Date:	

Code of Conduct
I, Vignesh P.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: 

Instructions

- This test contains 2 concept mapping. Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification, Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

Smart - Healthcare System

Data objects (real-entities / attributes) : Patient ID, Name, Age, Gender, Temp, BP, Diagnosis

Data Structure (way to store data)

Primitive Data Structure (single value data type) int, float, char, bool, double

Abstract Data Type (ADT)
(Define data + operation) Stack, Queue, List

Non-primitive data structure
Collection values, Arrays, Linked List, Trees.

Stack
LIFO - Last In First Out
→ used for undo operation

Operation
Enqueued()
Dequeued()
Front()
Is Empty

Queue
FIFO - First In First Out
→ used for patient flow management

Operation
Insert()
Delete()
Search()
Traversal

List
Linear Collection
→ used for storing medical records

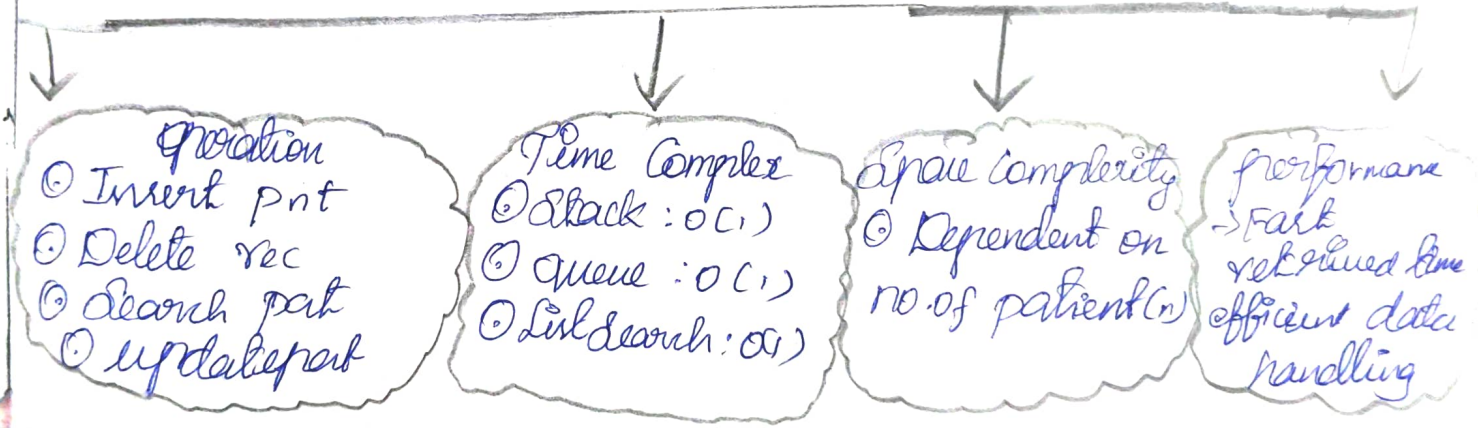
*)

Operations
push()
pop()
peek()
is Empty()

↓ Implemented using

Implementation In C - Language			
Structure struct patient int id; char name[50]; int age; float temp; };	Pointers used to link nodes in Linked List Tree struct node next;	Functions ① Insert() ① Delete() ① Search() ① Update() ① Display	Dynamic memory ① malloc() ① calloc() ① free()

↓ Leads to



Smart City Traffic Management

Data Structure
(The way of
organizing data)

Primitive Data
Structure
(Basic type of data)
int, float, char,
double, bool

Non-primitive Data
Structure
Made up of
collection of value
Array, Linked
List, Stack, Queue,
Tree

Example :
int (Veh-ID)
float (Speed)

Non-primitive Data
Structure

Linear Data Structure
Element arrangement
Segment

Non-Linear Data
Structure Element
in hierarchical
in network.

Array

→ Fixed Size
Of Similar
element
eg. Count[10]

Tree

→ Hierarchical
Structure
eg: Road hierarchy

Graph

→ Network ?
Node Connect by
edges

Linked List

→ Dynamic Collection
of elements
eg: Store List of
vehicles detect by
Sensors

Queue

FIFO
used for waiting
lines
eg: vehicles wait
at traffic signals.

Applications In Smart City Traffic

Vehicle
Monitoring

○ Store and
process vehicle
data from
Sensor

Traffic Signal
Control

○ Queue data
helps signal
Scheduling

Route
Optimization

Graph helps
find Shortest
/ fastest

Congest
Detection

Analyse
traffic
density
using array
& graph